

Interpretable Pulmonary-Nodule Malignancy Scoring

Prediction, WHERE, WHAT, WHY, and HOW

Author / ■■: [To be completed]

Affiliation / ■■: [To be completed]

Supervisor / ■■: [To be completed]

Date / ■■: 2026-08-13

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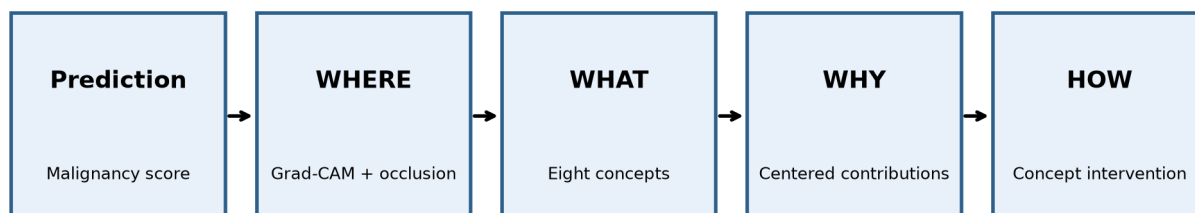
RPT-F13. Integrated Prediction-WHERE-WHAT-WHY-HOW interpretation and its boundaries.

Abbreviations

CBM — Concept Bottleneck Model; CEM — Concept Embedding Model; GAM — Generalized Additive Model; OOF — out-of-fold; ROI — region of interest; MAE — mean absolute error; RMSE — root mean squared error; AUROC — area under the receiver-operating-characteristic curve; AUPRC — area under the precision-recall curve.

Abstract

This study evaluates four three-dimensional deep-learning strategies for radiologist-assessed pulmonary-nodule malignancy using a frozen cohort of 2,633 nodules from 868 patients. The primary endpoint is pooled out-of-fold mean absolute error on the original 1–5 rating scale; secondary evidence covers the 1,073-nodule extreme subset, eight bottleneck concepts, centered score contributions, concept interventions, and spatial Grad-CAM faithfulness. The evidence is organised as Prediction, WHERE, WHAT, WHY, and HOW rather than as an audit inventory.



One frozen OOF evidence chain

RPT-F01. End-to-end evidence framework linking prediction to WHERE, WHAT, WHY, and HOW.

Learned-softmax GAM achieved the lowest primary point-estimate MAE of 0.480, compared with 0.501 for Black-box. Across all models, 73,724 Grad-CAM targets were requested: 66,769 were valid and 6,955 were explicitly recorded as post-ReLU all-zero maps. Matched-random occlusion showed that spatial saliency was not uniformly more faithful than random masks, while concept interventions produced model- and ordering-dependent changes.

The findings support a layered interpretation: accurate prediction does not by itself establish spatial or conceptual faithfulness; concept fidelity does not guarantee beneficial intervention; and additive contribution decompositions describe the model score without establishing clinical causality. Malignancy is a radiologist assessment rather than pathology-confirmed diagnosis, and the system is not a clinical diagnostic product.

1. Introduction

Pulmonary-nodule malignancy assessment combines a prediction problem with an explanation problem. A numeric malignancy score can be useful for benchmarking, yet a reader also needs to know where the image influenced the model, what radiological concepts the model represented, why those representations shifted the output, and how the output responds when concept information is corrected. Treating these as interchangeable forms of explanation obscures their different evidential roles.

We therefore frame the analysis around five linked questions. Prediction asks whether the continuous radiologist-assessed target is estimated accurately. WHERE uses Grad-CAM and matched occlusion to evaluate spatial sensitivity. WHAT measures the fidelity of six continuous and two categorical concept predictions. WHY decomposes concept-model scores into train-centered signed contributions and learned local-expert mixtures. HOW tests model dependence through preregistered concept interventions.

The contribution is not a new clinical classifier or a claim of pathology-level diagnosis. It is a controlled comparison of Black-box, Standard CBM, a project-specific Mixed-type CEM, and a preregistered Learned-softmax GAM under identical patient-grouped folds, shared encoder initialisations, exactly-once test evaluation, and a unified OOF analysis. The report keeps negative findings visible, including uncertain paired AUROC differences, limited

categorical concept fidelity, model-dependent intervention benefit, and concentrated zero-map behaviour.

2. Related Work

Three-dimensional convolutional networks provide a direct route from volumetric CT patches to malignancy scores, but their internal representations are not naturally expressed in radiological terms. Dense connectivity improves feature reuse and gradient flow, motivating the shared DenseNet-121 encoder used across the four models. The Black-box model establishes what this image-first route can achieve without concept constraints.

RPT-T01. Related-work comparison

Approach	Prediction	Concepts	Spatial explanation	Intervention	This study
Black-box CNN	Yes	No	Optional	No	Comparator
Concept Bottleneck Model	Yes	Explicit	Concept/task Grad-CAM	Concept replacement	Standard CBM
Concept Embedding Model	Yes	Mixed-type embeddings	Concept/task Grad-CAM	Mixture-weight replacement	Project-specific Mixed-type CEM
Additive local experts	Yes	Explicit	Concept/task Grad-CAM	Local-expert re-evaluation	Preregistered Learned-softmax GAM

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T01.csv.

Concept bottleneck models expose intermediate variables that can be evaluated and intervened upon. Concept embedding models relax a scalar bottleneck by constructing sample-conditioned concept states, but this project extends that idea to mixed continuous and categorical reader targets. The Learned-softmax GAM instead preserves explicit concept predictions and sends each group through five local experts whose mixture weights are learned at fold level.

Grad-CAM localises gradients at a convolutional feature layer, but a visually concentrated heatmap is not automatically faithful. This study therefore compares deterministic saliency masks with 20 equal-size random masks and preserves both `output_sensitivity` and `error_increase`. The distinction is essential: output movement is not evidence that the prediction became worse. Table RPT-T01 situates these components without presenting prior-work results as evidence for the present cohort.

3. Dataset and Preprocessing

The study uses the LIDC-IDRI XML reader annotations and stable physical-nodule identities. The frozen primary cohort contains 2,633 nodules from 868 patients; 1,073 nodules from 578 patients satisfy the preregistered extreme definition, with 782 low and 291 high cases. Patient Diagnoses XLS is not used for training, and malignancy is the mean of valid radiologist ratings rather than a pathology-confirmed label.

RPT-T02. Frozen cohort flow

Cohort component	Nodules	Patients	Role
Reference physical nodules	2651	875	Reconciliation only
Missing required target	1	NA	Excluded
Primary regression	2633	868	Main five-fold evaluation
Secondary extreme subset	1073	578	782 low / 291 high

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T02.csv.

Malignancy is the downstream 1–5 target and is not one of the eight bottleneck concepts. The concepts are subtlety, internalStructure, calcification, sphericity, margin, lobulation, spiculation, and texture. Six targets are continuous normalized reader means; internalStructure and calcification retain complete reader vote distributions, including true modal ties for training and soft metrics.

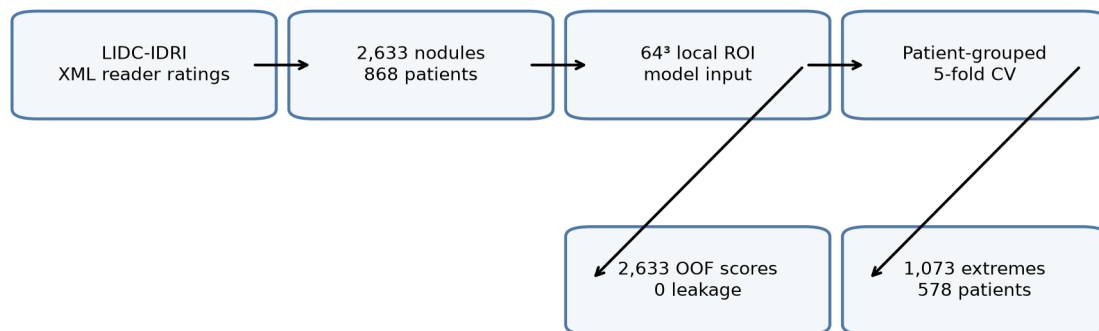
RPT-T03. Target and concept definitions

Variable	Role	Type	Frozen target
Malignancy	Downstream target (not a concept)	Continuous	Radiologist mean, 1–5; normalized $(y-1)/4$
subtlety	Bottleneck concept	Continuous	Normalized valid-reader mean
sphericity	Bottleneck concept	Continuous	Normalized valid-reader mean
margin	Bottleneck concept	Continuous	Normalized valid-reader mean
lobulation	Bottleneck concept	Continuous	Normalized valid-reader mean
spiculation	Bottleneck concept	Continuous	Normalized valid-reader mean
texture	Bottleneck concept	Continuous	Normalized valid-reader mean
internalStructure	Bottleneck concept	Categorical (4 classes)	Full reader vote distribution
calcification	Bottleneck concept	Categorical (6 classes)	Full reader vote distribution

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T03.csv.

Each model receives a $64 \times 64 \times 64$ local pulmonary-nodule ROI created by consensus-mask cropping, cubic padding, and deterministic resampling. The ROI is not a complete axial CT slice and can appear lower-resolution because it has been cropped and resampled. Full axial CT is used only as private contextual visualization when exact frozen series, slice, bounding-box, and coordinate provenance are available. Figure RPT-F02 and Tables RPT-T02–RPT-T03 show the study-specific cohort and variables.

Cohort, preprocessing, and evaluation flow



RPT-F02. Frozen cohort, local ROI preprocessing, and patient-grouped five-fold evaluation flow.

4. Methods

All four models share the same fold-specific DenseNet-121 encoder initialisation and use an unconstrained linear malignancy output. Scores are trained and evaluated without sigmoid, tanh, or clipping. Black-box maps encoder features directly to the score. Standard CBM first learns the eight concepts and then fits a linear task head using frozen predicted concepts. Mixed-type CEM forms sample-conditioned states for continuous and categorical concepts. Learned-softmax GAM applies five local experts to each predicted concept group and adds their softmax-weighted outputs.

RPT-T04. Four-model architecture comparison

Model	Task path	Concept representation	Contribution semantics	Intervention semantics
Black-box	DenseNet features → linear score	None	Not applicable	Not applicable
Standard CBM	Predicted concepts → linear score	6 sigmoid + 2 softmax groups	Linear group terms	Replace activated concept group
Mixed-type CEM	Sample-conditioned concept embeddings → linear score	Mixed-type dynamic states	Embedding block dot product	Replace mixture weights only

Model	Task path	Concept representation	Contribution semantics	Intervention semantics
Learned-softmax GAM	Predicted concepts → local experts → additive score	6 sigmoid + 2 softmax groups	Softmax-weighted local experts	Ground-truth concept through experts

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T04.csv.

Concept-model contributions are centred using means computed only from the current training fold. The centered bias plus eight centered contributions reconstructs the normalized score, and multiplying contributions by 4 reconstructs the original rating-point scale. These signed terms describe how the trained model composes its output; centering constants are bookkeeping statistics, not feature importance. The unavailable mean absolute aggregate is not recreated for presentation.

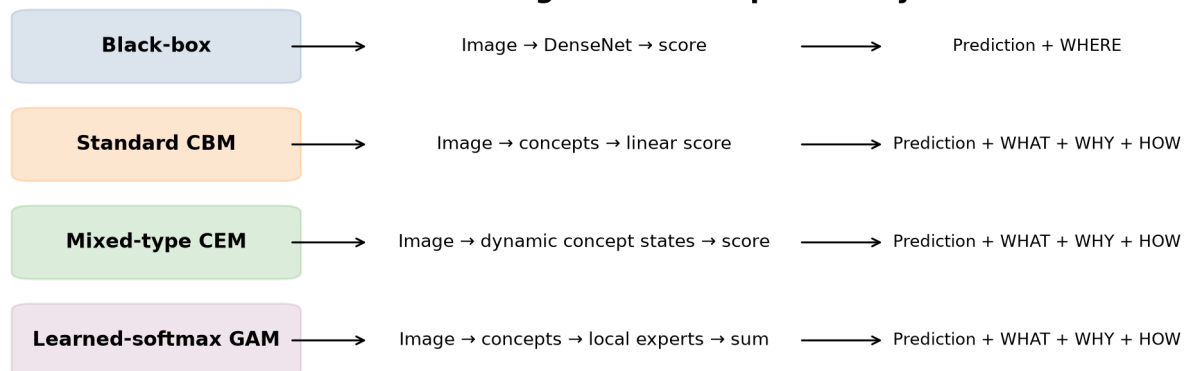
RPT-T06. Evaluation protocol

Component	Unit	Metric	Selection/uncertainty
Primary regression	Nodule; patient-cluster bootstrap	Unclipped original-scale MAE (primary), RMSE, normalized MAE, Pearson, Spearman	2,000 shared patient draws
Secondary extreme	1,073 extreme nodules / 578 patients	AUROC, AUPRC; threshold metrics	Fold-validation extreme-only Youden-J; 2,000 valid draws
Concept fidelity	Nodule	Continuous MAE/RMSE/correlation; categorical CE/Brier/macro-F1	Hard F1 excludes true modal ties
Spatial faithfulness	Valid Grad-CAM target	output_sensitivity and error_increase	26,215 voxels; 20 matched random masks
Intervention	Pooled OOF	iMAE/Delta_iMAE; iAUC/Delta_iAUC	k=0...8; random and error-first orderings

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T06.csv.

Grad-CAM uses the final registered convolutional layer, spatial-mean gradients, a weighted activation sum, ReLU, and trilinear upsampling to 64³. Maps remain raw FP32 scientific artifacts. Display overlays may be normalized only for visualization. A zero post-ReLU map is marked undefined and excluded from the occlusion denominator; the frozen artifacts do not contain the pre-ReLU, gradient-norm, activation-norm, or channel-weight decomposition required to infer its exact mechanism.

Four architectures and registered interpretability interfaces



RPT-F03. Four model architectures and their registered interpretability interfaces.

Occlusion replaces the top 26,215 heatmap voxels with normalized zero and compares them with 20 uniform-without-replacement random masks of equal size. output_sensitivity is the absolute output movement. error_increase is the change in absolute target error and is positive only when prediction error worsens. Intervention curves replace 0...8 concept groups under shared random permutations or error-first ordering; positive Delta_iMAE and Delta_iAUC consistently denote improvement. Figure RPT-F03 and Tables RPT-T04 and RPT-T06 summarise these registered semantics.

5. Experimental Setup

Evaluation uses patient-grouped five-fold outer cross-validation with fixed test counts of 479, 502, 539, 549, and 564 nodules. Patients are disjoint within each fold partition, and every primary nodule appears exactly once in the canonical OOF test set. Fold-specific validation subsets select checkpoints and Youden-J thresholds; test labels never enter selection.

RPT-T05. Frozen training configuration and completion

Model	Fold	Best epoch	Epochs complete	Test transactions	Scientific status
Black-box	0	14	NA	1	PASS
Black-box	1	19	NA	1	PASS
Black-box	2	10	NA	1	PASS
Black-box	3	15	NA	1	PASS
Black-box	4	38	NA	1	PASS
Standard CBM	0	concept=10;task=78	NA	1	PASS
Standard CBM	1	concept=16;task=79	NA	1	PASS
Standard CBM	2	concept=38;task=77	NA	1	PASS
Standard CBM	3	concept=6;task=79	NA	1	PASS
Standard CBM	4	concept=18;task=78	NA	1	PASS
Mixed-type CEM	0	28	NA	1	PASS
Mixed-type CEM	1	21	NA	1	PASS
Mixed-type CEM	2	18	NA	1	PASS
Mixed-type CEM	3	15	NA	1	PASS
Mixed-type CEM	4	44	NA	1	PASS
Learned-softmax GAM	0	10	NA	1	PASS
Learned-softmax GAM	1	28	NA	1	PASS
Learned-softmax GAM	2	32	NA	1	PASS
Learned-softmax GAM	3	15	NA	1	PASS
Learned-softmax GAM	4	22	NA	1	PASS
Black-box	None	NA	NA	1	PASS
Learned-softmax GAM	None	NA	NA	1	PASS
Mixed-type CEM	None	NA	NA	1	PASS
Standard CBM	None	NA	NA	1	PASS

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T05.csv.

Each model/fold training run uses one frozen seed and at most 80 epochs. Test evaluation is committed exactly once after the best checkpoint is fixed. Historical scheduler failures and verifier recoveries are retained as provenance but are distinguished from scientific validity; completed histories, checkpoints, predictions, metrics, and evaluations are never overwritten by reporting.

RPT-T06. Evaluation protocol

Component	Unit	Metric	Selection/uncertainty
Primary regression	Nodule; patient-cluster bootstrap	Unclipped original-scale MAE (primary), RMSE, normalized MAE, Pearson, Spearman	2,000 shared patient draws
Secondary extreme	1,073 extreme nodules / 578 patients	AUROC, AUPRC; threshold metrics	Fold-validation extreme-only Youden-J; 2,000 valid draws
Concept fidelity	Nodule	Continuous MAE/RMSE/correlation; categorical CE/Brier/macro-F1	Hard F1 excludes true modal ties
Spatial faithfulness	Valid Grad-CAM target	output_sensitivity and error_increase	26,215 voxels; 20 matched random masks
Intervention	Pooled OOF	iMAE/Delta_iMAE; iAUC/Delta_iAUC	k=0...8; random and error-first orderings

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T06.csv.

Uncertainty uses 2,000 patient-cluster bootstrap replicates, with shared patient draws for paired comparisons. Each selected patient carries all of their nodules. Secondary AUROC draws are redrawn when they contain a single class. Table RPT-T05 records fold-level completion, while Table RPT-T06 separates statistical definitions from execution history.

6.1 Results — Prediction

What was measured? Primary prediction was evaluated on all 2,633 OOF nodules with original-scale MAE as the primary endpoint. Learned-softmax GAM produced the lowest point estimate (0.480), followed by Mixed-type CEM (0.484), Black-box (0.501), and Standard CBM (0.502). Table RPT-T07 reports every frozen regression point estimate and its existing 2,000-draw interval; Figure RPT-F04 makes the overlap in uncertainty visible.

RPT-T07. Primary regression

Model	MAE (95% CI)	RMSE (95% CI)	Normalized MAE (95% CI)	Pearson (95% CI)	Spearman (95% CI)	Prediction range (1–5)	N
Black-box	0.501 (0.483–0.520)	0.642 (0.619–0.667)	0.125 (0.121–0.130)	0.716 (0.689–0.741)	0.635 (0.599–0.668)	0.489–5.120	2633
Learned-softmax GAM	0.480 (0.462–0.498)	0.618 (0.592–0.642)	0.120 (0.116–0.125)	0.741 (0.712–0.768)	0.653 (0.616–0.688)	1.008–4.682	2633
Mixed-type CEM	0.484 (0.467–0.502)	0.628 (0.604–0.654)	0.121 (0.117–0.126)	0.730 (0.701–0.757)	0.640 (0.604–0.673)	0.823–4.935	2633
Standard CBM	0.502 (0.483–0.522)	0.650 (0.625–0.675)	0.126 (0.121–0.131)	0.708 (0.677–0.735)	0.609 (0.570–0.648)	0.858–4.580	2633

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T07.csv.

What did we observe? Paired Delta-MAE supports Learned-softmax GAM over Black-box and Standard CBM because the corresponding intervals do not cross zero, whereas smaller differences require a more cautious reading. The Black-box versus Standard CBM interval crosses zero, showing that interpretability structure did not automatically improve point prediction. Table RPT-T08 and Figure RPT-F05 preserve all six comparisons and the sign convention $MAE_A - MAE_B$.

RPT-T08. Six paired Delta-MAE comparisons

Comparison (A vs B)	Delta-MAE (A–B)	95% CI	Crosses zero	Direction
Black-box vs Learned-softmax GAM	0.020	0.010–0.031	False	Positive supports B
Black-box vs Mixed-type CEM	0.016	0.006–0.027	False	Positive supports B
Black-box vs Standard CBM	-0.002	-0.015–0.012	True	Positive supports B
Mixed-type CEM vs Learned-softmax GAM	0.004	-0.006–0.013	True	Positive supports B
Standard CBM vs Learned-softmax GAM	0.022	0.011–0.033	False	Positive supports B
Standard CBM vs Mixed-type CEM	0.018	0.006–0.030	False	Positive supports B

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T08.csv.

On the 1,073-nodule extreme subset, all four continuous scores discriminated low from high ratings, but paired Delta-AUROC evidence was less decisive than the MAE evidence. Several intervals cross zero, and Standard CBM is lower than Black-box under the registered B–A convention. Table RPT-T09, Table RPT-T10, and Figure RPT-F06 therefore separate absolute AUROC/AUPRC performance from between-model uncertainty.

RPT-T09. Extreme-task performance

Model	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity	Specificity	Balanced accuracy	N
Black-box	0.945 (0.926–0.962)	0.894 (0.859–0.925)	0.801	0.927	0.864	1073

Model	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity	Specificity	Balanced accuracy	N
Learned-softmax GAM	0.949 (0.927–0.968)	0.903 (0.868–0.934)	0.859	0.894	0.876	1073
Mixed-type CEM	0.942 (0.920–0.960)	0.877 (0.833–0.916)	0.832	0.923	0.877	1073
Standard CBM	0.933 (0.911–0.951)	0.866 (0.826–0.900)	0.825	0.866	0.845	1073

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T09.csv.

What does this mean? Learned-softmax GAM is the strongest point-estimate regressor in this experiment, but the result does not justify a universal ranking across endpoints. Unclipped score ranges and small out-of-range rates remain part of the model behaviour rather than being hidden by post-hoc clipping. The target is a radiologist mean, so predictive accuracy should not be interpreted as pathology-level diagnostic accuracy.

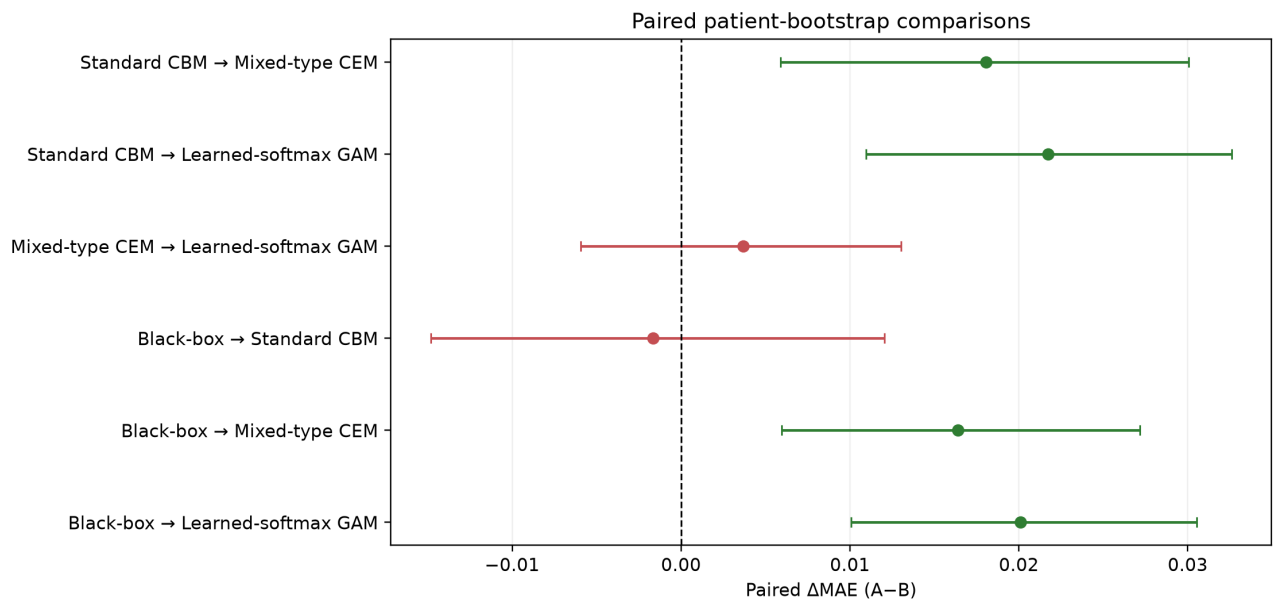
RPT-T10. Six paired Delta-AUROC comparisons

Comparison (A vs B)	Delta-AUROC (B–A)	95% CI	Crosses zero	Direction
Black-box vs Learned-softmax GAM	0.004	-0.005–0.014	True	Positive supports B
Black-box vs Mixed-type CEM	-0.004	-0.013–0.005	True	Positive supports B
Black-box vs Standard CBM	-0.013	-0.022–0.004	False	Positive supports B
Mixed-type CEM vs Learned-softmax GAM	0.008	-0.001–0.018	True	Positive supports B
Standard CBM vs Learned-softmax GAM	0.017	0.006–0.027	False	Positive supports B
Standard CBM vs Mixed-type CEM	0.009	-0.002–0.020	True	Positive supports B

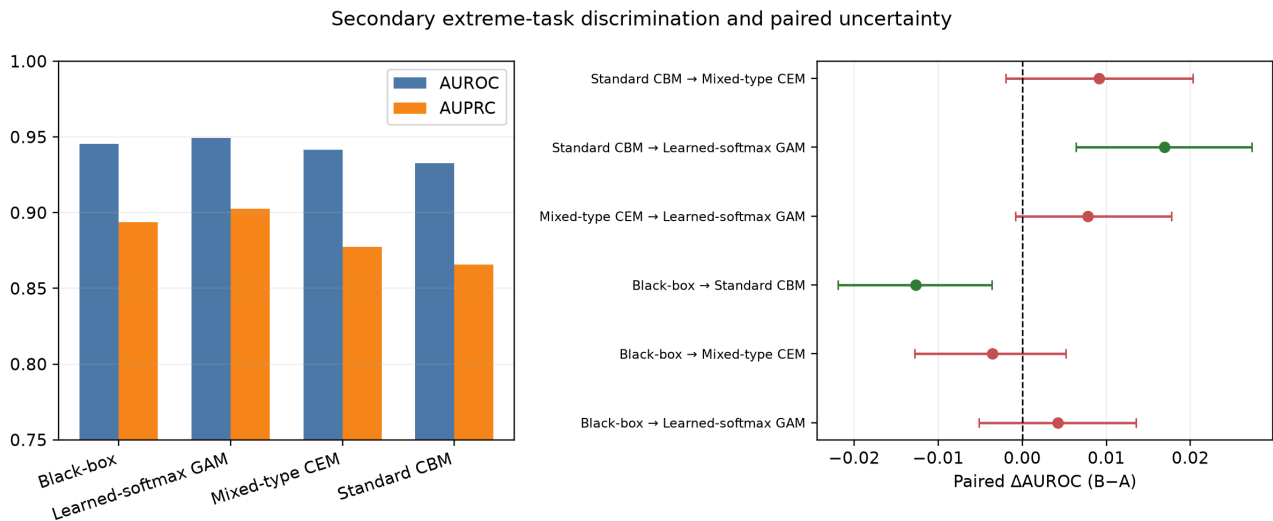
Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T10.csv.



RPT-F04. Pooled primary MAE with 2,000 patient-cluster bootstrap 95% intervals.



RPT-F05. Six paired Delta-MAE comparisons; intervals crossing zero are shown separately.



RPT-F06. Extreme-task AUROC/AUPRC and six paired Delta-AUROC comparisons.

Scientific conclusion codes: GAM_LOWEST_POINT_ESTIMATE_MAE, PAIRED_MAE_SUPPORTS_GAM_OVER_BLACKBOX_AND_CBM, AUROC_DIFFERENCES_MOSTLY_UNCERTAIN

6.2 Results — WHERE

What was measured? Spatial evidence comprises 73,724 requested Grad-CAM maps across all model, fold, and target combinations. Exactly 66,769 maps were valid and 6,955 were post-ReLU all-zero, yielding an overall undefined rate of 9.434%. Table RPT-T13 provides full accounting, while Figure RPT-F07 reveals concentrations that a pooled count would conceal.

RPT-T13. Grad-CAM accounting

Model	Fold	Target	Requested	Valid	Undefined	Undefined rate
Mixed-type CEM	0	malignancy	479	102	377	0.787
Mixed-type CEM	4	internalStructure	564	167	397	0.704
Mixed-type CEM	0	internalStructure	479	170	309	0.645
Mixed-type CEM	4	malignancy	564	203	361	0.640
Learned-softmax GAM	3	calcification	549	198	351	0.639

Model	Fold	Target	Requested	Valid	Undefined	Undefined rate
Standard CBM	0	calcification	479	183	296	0.618
Mixed-type CEM	2	malignancy	539	213	326	0.605
Mixed-type CEM	1	calcification	502	202	300	0.598
Standard CBM	1	calcification	502	309	193	0.384
Mixed-type CEM	1	malignancy	502	318	184	0.367
Mixed-type CEM	4	calcification	564	368	196	0.348
Mixed-type CEM	2	internalStructure	539	362	177	0.328
Learned-softmax GAM	2	spiculation	539	367	172	0.319
Mixed-type CEM	4	sphericity	564	384	180	0.319
Standard CBM	4	calcification	564	405	159	0.282
Mixed-type CEM	0	calcification	479	356	123	0.257
Learned-softmax GAM	3	spiculation	549	410	139	0.253
Mixed-type CEM	2	lobulation	539	405	134	0.249
Learned-softmax GAM	4	calcification	564	435	129	0.229
Learned-softmax GAM	3	lobulation	549	426	123	0.224
Learned-softmax GAM	3	malignancy	549	426	123	0.224
Mixed-type CEM	3	malignancy	549	431	118	0.215
Standard CBM	0	internalStructure	479	382	97	0.203
Black-box	3	malignancy	549	448	101	0.184

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T13.csv.

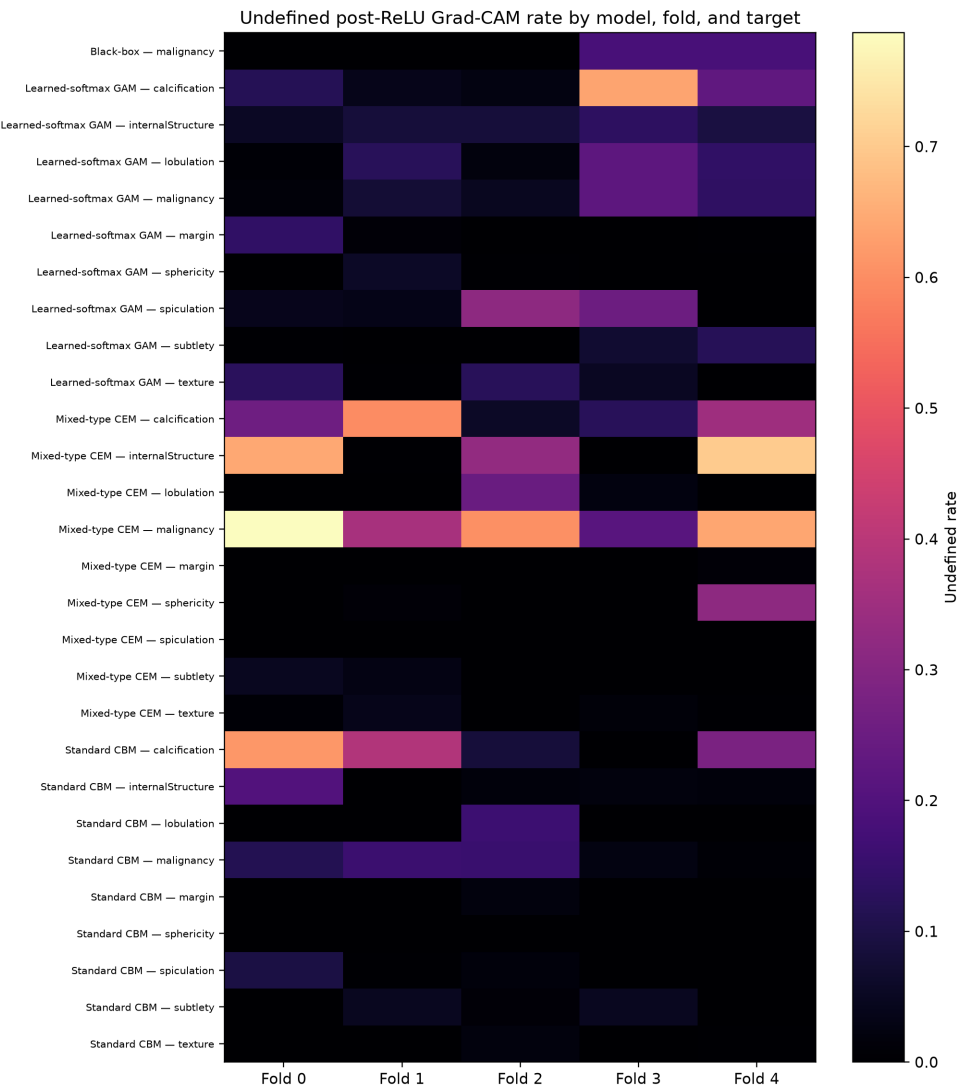
What did we observe? Undefined maps were not uniformly distributed. They were concentrated in particular model-target combinations, which is why the frozen root-cause label is `SYSTEMATIC_MODEL/TARGET_ISSUE` rather than an undifferentiated implementation failure. Every undefined map was finite and exactly zero after ReLU; no NaN, Inf, loading error, or target-path mismatch was observed. However, the persisted artifacts do not contain pre-ReLU CAMs or gradient/channel-weight norms.

RPT-T14. Spatial faithfulness

Model	Target	Quantity	Saliency mean	Saliency median	Saliency-random mean	Saliency > random rate	Valid maps
Black-box	malignancy	output_sensitivity	0.025	0.017	-0.323	0.009	2429
Black-box	malignancy	error_increase	0.003	0.000	-0.235	0.144	2429
Learned-softmax GAM	malignancy	output_sensitivity	0.025	0.020	-0.302	0.004	2361
Learned-softmax GAM	malignancy	error_increase	0.002	0.000	-0.214	0.157	2361
Mixed-type CEM	malignancy	output_sensitivity	0.026	0.020	-0.318	0.007	1267
Mixed-type CEM	malignancy	error_increase	0.005	0.003	-0.247	0.154	1267
Standard CBM	malignancy	output_sensitivity	0.018	0.012	-0.218	0.005	2398
Standard CBM	malignancy	error_increase	0.003	0.001	-0.140	0.227	2398

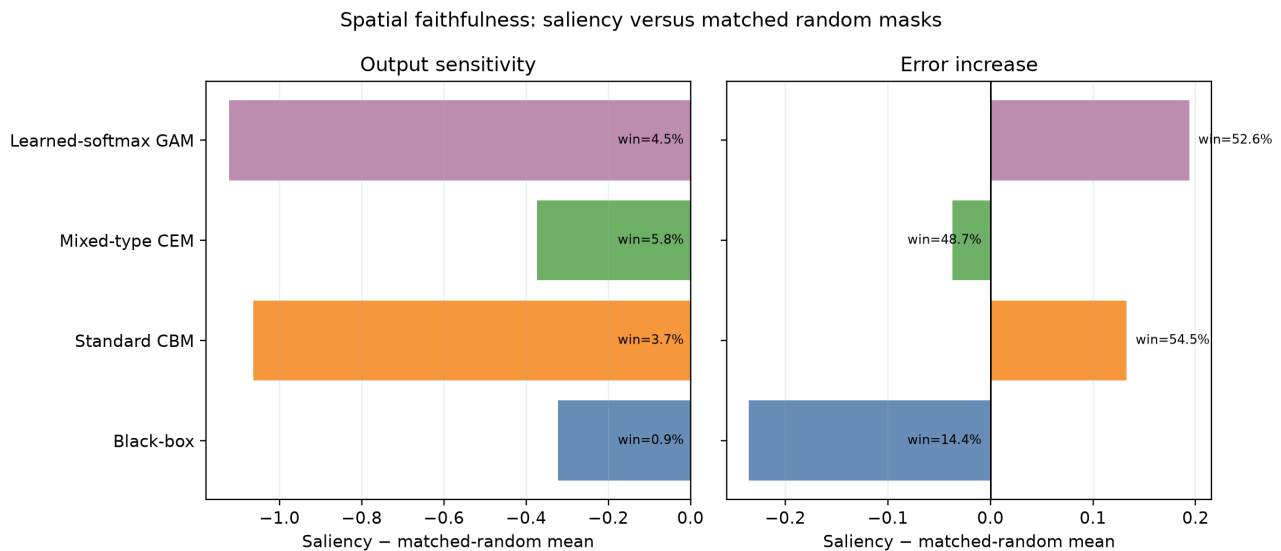
Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T14.csv.

Faithfulness produced a scientifically important negative result. For both `output_sensitivity` and `error_increase`, pooled saliency-minus-random means were often negative, and saliency exceeded the matched random mean in only a minority of valid cases. Table RPT-T14 and Figure RPT-F08 keep output movement separate from prediction-error worsening. A large `output_sensitivity` cannot by itself show that the prediction became less accurate.



RPT-F07. Undefined post-ReLU Grad-CAM rate by model, fold, and target.

What does this mean? Grad-CAM provides a spatial sensitivity proxy, not a ground-truth localisation claim. The zero-map concentration and weak matched-random advantage limit strong spatial interpretations even when the underlying task prediction is accurate. Display overlays are normalized only for qualitative reading; every quantitative occlusion result uses the original unnormalized FP32 map.



RPT-F08. Spatial faithfulness for output_sensitivity and error_increase versus matched random masks.

Scientific conclusion codes: SALIENCY_NOT_UNIFORMLY_MORE_FAITHFUL_THAN_RANDOM,
SYSTEMATIC_MODEL_TARGET_ZERO_MAP_LIMITATION

6.3 Results — WHAT

What was measured? Continuous concept fidelity uses MAE, RMSE, Pearson, and Spearman over 2,633 nodules for each concept model. Categorical fidelity uses soft cross-entropy and multiclass Brier on the full reader-vote distributions, plus hard modal macro-F1 only where the true modal class is unique. Table RPT-T11 and Figure RPT-F09A keep continuous metrics on compatible scales; Table RPT-T12 and Figure RPT-F09B do the same for categorical evidence.

RPT-T11. Continuous concept metrics

Model	Concept	MAE	RMSE	Pearson	Spearman	N
Learned-softmax GAM	lobulation	0.127	0.190	0.451	0.472	2633
Learned-softmax GAM	margin	0.145	0.198	0.683	0.648	2633
Learned-softmax GAM	sphericity	0.140	0.178	0.424	0.435	2633
Learned-softmax GAM	spiculation	0.114	0.184	0.499	0.456	2633
Learned-softmax GAM	subtlety	0.168	0.221	0.578	0.582	2633
Learned-softmax GAM	texture	0.111	0.179	0.786	0.583	2633
Mixed-type CEM	lobulation	0.183	0.218	0.270	0.249	2633
Mixed-type CEM	margin	0.246	0.292	0.188	0.171	2633
Mixed-type CEM	sphericity	0.178	0.217	0.056	0.051	2633
Mixed-type CEM	spiculation	0.180	0.216	0.305	0.274	2633
Mixed-type CEM	subtlety	0.204	0.250	0.445	0.449	2633
Mixed-type CEM	texture	0.265	0.305	0.202	0.167	2633
Standard CBM	lobulation	0.131	0.186	0.460	0.456	2633
Standard CBM	margin	0.146	0.197	0.687	0.647	2633
Standard CBM	sphericity	0.139	0.177	0.437	0.453	2633
Standard CBM	spiculation	0.121	0.182	0.507	0.416	2633
Standard CBM	subtlety	0.169	0.223	0.567	0.573	2633
Standard CBM	texture	0.116	0.180	0.783	0.581	2633

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T11.csv.

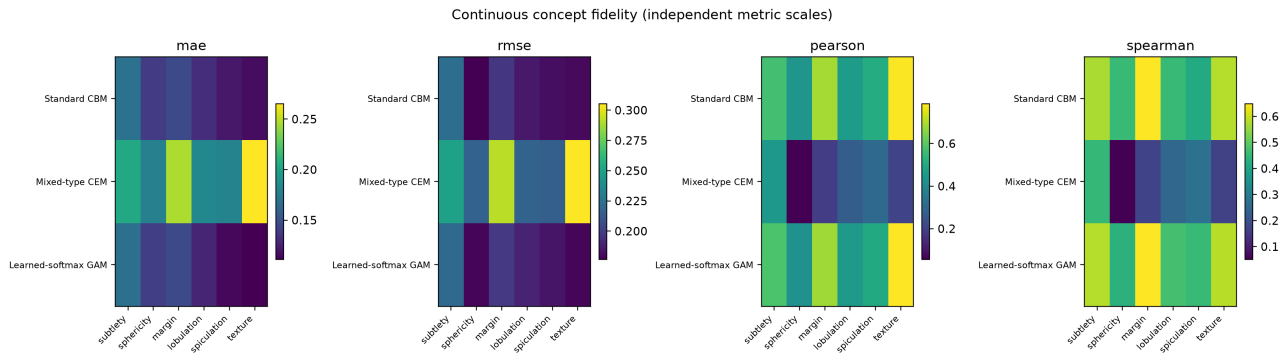
What did we observe? Continuous fidelity differed substantially by concept and model rather than following a single model-wide pattern. Some morphological concepts showed useful correlation, while subtle or reader-variable concepts retained larger absolute errors. This heterogeneity matters because the downstream concept models can only explain their own predicted representations, not an error-free radiological state.

RPT-T12. Categorical concept metrics

Model	Concept	Soft CE	Brier	Macro-F1	Soft N	Hard N	Ties
Learned-softmax GAM	calcification	0.201	0.048	0.313	2633	2578	55
Learned-softmax GAM	internalStructure	0.038	0.007	0.312	2633	2625	8
Mixed-type CEM	calcification	0.262	0.068	0.310	2633	2578	55
Mixed-type CEM	internalStructure	0.083	0.014	0.250	2633	2625	8
Standard CBM	calcification	0.207	0.049	0.314	2633	2578	55
Standard CBM	internalStructure	0.039	0.007	0.250	2633	2625	8

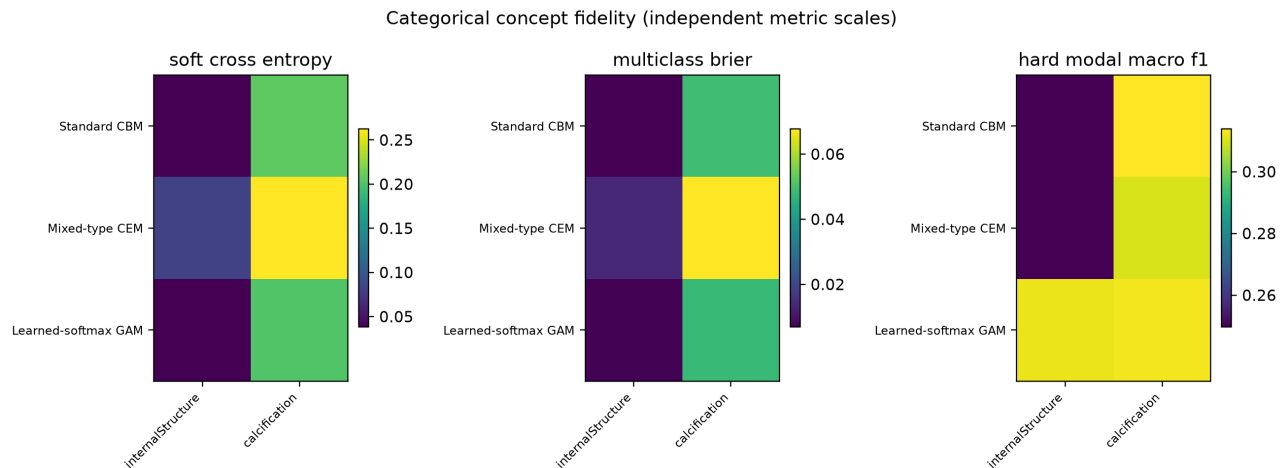
Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T12.csv.

Categorical results were more limited. internalStructure and calcification retain complete vote distributions, so soft losses and Brier scores are the authoritative distributional evidence. Hard modal macro-F1 is included for readability but excludes true ties and can be low when rare classes are difficult. Treating the modal label as a single expert ground truth would misstate the frozen target.



RPT-F09A. Continuous concept fidelity on independent metric scales.

What does this mean? WHAT evidence supports inspecting individual concept groups rather than declaring that a model has uniformly learned radiological concepts. Concept fidelity is necessary for a transparent bottleneck but is not sufficient for predictive superiority or intervention benefit. The private RPT-TA02 table therefore presents both continuous targets and categorical vote-distribution semantics at case level.



RPT-F09B. Categorical concept fidelity on independent metric scales.

6.4 Results — WHY

What was measured? WHY evidence asks how predicted concepts enter each concept model's malignancy score. For every fold, train-only means centre the raw group terms, and the centered bias plus eight terms reconstructs the task score within the frozen 1e-6 tolerance. Table RPT-T15 records pooled signed means and fold centering constants without relabelling them as importance.

RPT-T15. Centered contribution summary

Model	Concept	Pooled signed mean (rating points)	Fold train-centering constants (rating points)	Mean [contribution]	Interpretation
Learned-softmax GAM	calcification	0.402	0.450; 0.247; 0.397; 0.498; 0.419	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	internalStructure	0.041	-0.275; 0.289; 0.032; -0.160; 0.321	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	lobulation	-0.016	0.336; 0.133; -0.834; -0.039; 0.325	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	margin	0.370	0.526; 0.290; 0.647; 0.226; 0.162	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	sphericity	0.106	-0.127; 0.055; 0.449; 0.109; 0.045	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	spiculation	0.323	0.058; 0.178; 0.502; 0.611; 0.265	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	subtlety	0.229	0.254; 0.172; 0.257; 0.160; 0.299	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Learned-softmax GAM	texture	0.190	0.355; 0.351; 0.181; 0.275; -0.209	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	calcification	0.263	0.215; 0.511; 0.245; 0.201; 0.146	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	internalStructure	0.486	0.371; 0.153; 0.585; 0.473; 0.848	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	lobulation	0.279	0.321; 0.155; 0.217; 0.390; 0.310	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	margin	0.123	0.219; 0.046; 0.007; 0.338; 0.006	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance

Model	Concept	Pooled signed mean (rating points)	Fold train-centering constants (rating points)	Mean contribution	Interpretation
Mixed-type CEM	sphericity	0.155	0.440; 0.057; 0.188; 0.090; -0.001	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	spiculation	0.141	0.042; 0.195; 0.213; 0.112; 0.145	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	subtlety	0.133	-0.039; 0.185; 0.298; 0.091; 0.127	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Mixed-type CEM	texture	0.101	0.147; 0.221; 0.044; 0.045; 0.050	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	calcification	0.400	0.941; 0.633; 0.387; 0.145; -0.104	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	internalStructure	0.281	-0.005; 0.479; 0.482; -0.254; 0.704	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	lobulation	0.246	0.260; 0.129; 0.211; 0.243; 0.385	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	margin	0.137	0.152; -0.041; -0.131; 0.397; 0.310	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	sphericity	-0.072	0.482; 0.104; -0.251; -0.334; -0.362	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	spiculation	0.159	0.227; 0.231; 0.080; 0.144; 0.113	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	subtlety	0.285	0.038; -0.027; 0.793; 0.559; 0.063	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance
Standard CBM	texture	-0.122	-0.155; 0.261; -0.227; -0.349; -0.137	DATA_NOT_PERSISTED	Signed centering evidence; constants are not importance

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T15.csv.

What did we observe? Signed contribution directions differ across concept and model, demonstrating that identical concept names need not play identical decision roles. Figure RPT-F10 displays empirical OOF profiles derived as a presentation summary of frozen per-sample points. The profiles are descriptive and should not be read as global causal shape functions. The authoritative model-by-concept mean absolute aggregate was not persisted, so the report marks it DATA_NOT_PERSISTED rather than recreating it.

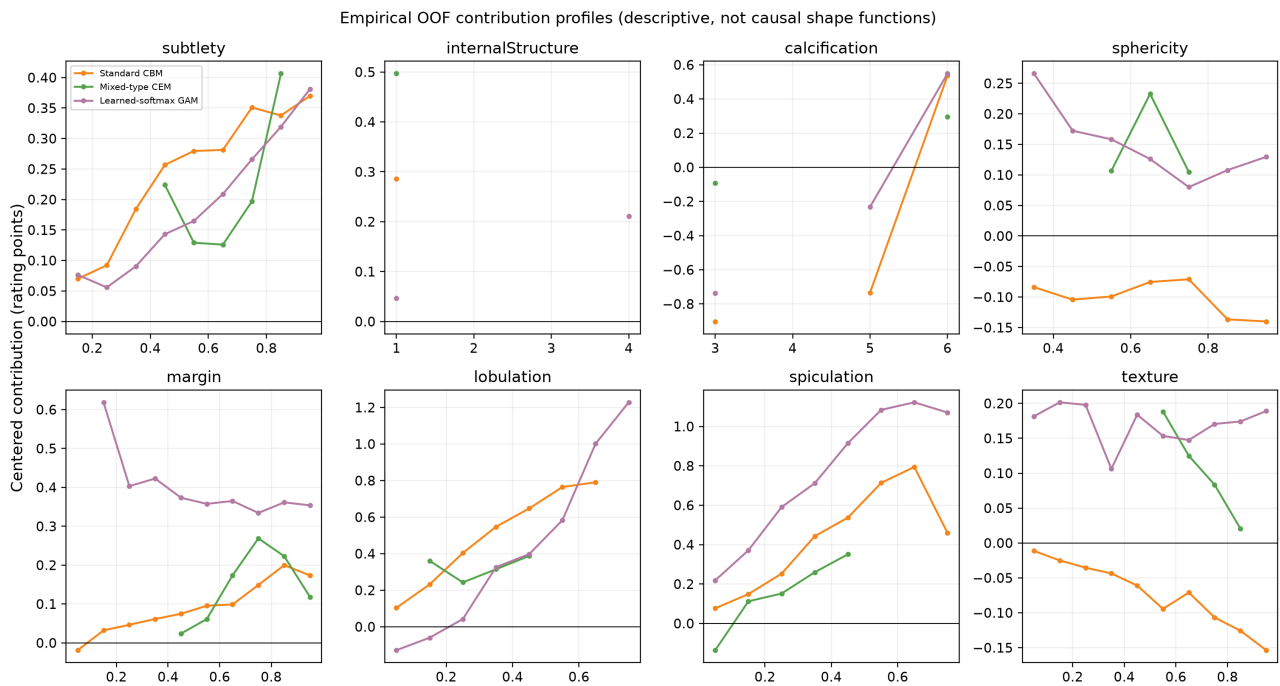
RPT-T16. Fold-level Learned-softmax GAM alpha

Fold	Concept	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5	Min-max
0	calcification	0.200	0.200	0.204	0.196	0.199	0.196-0.204
0	internalStructure	0.200	0.200	0.200	0.200	0.200	0.200-0.200
0	lobulation	0.201	0.203	0.198	0.200	0.197	0.197-0.203
0	margin	0.198	0.198	0.203	0.202	0.200	0.198-0.203
0	sphericity	0.199	0.200	0.200	0.202	0.200	0.199-0.202
0	spiculation	0.194	0.200	0.200	0.200	0.207	0.194-0.207
0	subtlety	0.206	0.195	0.198	0.202	0.198	0.195-0.206
0	texture	0.201	0.200	0.199	0.200	0.200	0.199-0.201
1	calcification	0.195	0.191	0.201	0.201	0.211	0.191-0.211
1	internalStructure	0.201	0.199	0.199	0.199	0.201	0.199-0.201
1	lobulation	0.197	0.199	0.203	0.206	0.195	0.195-0.206
1	margin	0.202	0.200	0.199	0.199	0.201	0.199-0.202
1	sphericity	0.199	0.199	0.201	0.199	0.202	0.199-0.202

Fold	Concept	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5	Min-max
1	spiculation	0.200	0.203	0.205	0.195	0.197	0.195–0.205
1	subtlety	0.203	0.200	0.196	0.201	0.200	0.196–0.203
1	texture	0.202	0.200	0.199	0.198	0.201	0.198–0.202
2	calcification	0.200	0.196	0.206	0.199	0.199	0.196–0.206
2	internalStructure	0.199	0.199	0.201	0.201	0.201	0.199–0.201
2	lobulation	0.196	0.202	0.206	0.192	0.204	0.192–0.206
2	margin	0.201	0.199	0.198	0.200	0.202	0.198–0.202
2	sphericity	0.200	0.202	0.199	0.199	0.201	0.199–0.202
2	spiculation	0.207	0.201	0.202	0.196	0.193	0.193–0.207
2	subtlety	0.198	0.201	0.202	0.198	0.201	0.198–0.202
2	texture	0.201	0.199	0.199	0.199	0.201	0.199–0.201
3	calcification	0.198	0.203	0.200	0.196	0.202	0.196–0.203
3	internalStructure	0.201	0.200	0.199	0.199	0.201	0.199–0.201
3	lobulation	0.195	0.214	0.195	0.197	0.200	0.195–0.214
3	margin	0.200	0.203	0.199	0.200	0.198	0.198–0.203
3	sphericity	0.201	0.199	0.200	0.201	0.198	0.198–0.201
3	spiculation	0.203	0.203	0.192	0.200	0.201	0.192–0.203
3	subtlety	0.198	0.202	0.200	0.199	0.201	0.198–0.202
3	texture	0.200	0.199	0.200	0.201	0.199	0.199–0.201
4	calcification	0.203	0.206	0.195	0.193	0.202	0.193–0.206
4	internalStructure	0.200	0.201	0.199	0.201	0.199	0.199–0.201
4	lobulation	0.190	0.203	0.204	0.206	0.198	0.190–0.206
4	margin	0.201	0.200	0.199	0.200	0.201	0.199–0.201
4	sphericity	0.199	0.199	0.202	0.199	0.201	0.199–0.202
4	spiculation	0.198	0.201	0.198	0.203	0.199	0.198–0.203
4	subtlety	0.196	0.200	0.200	0.203	0.201	0.196–0.203
4	texture	0.199	0.199	0.200	0.202	0.200	0.199–0.202

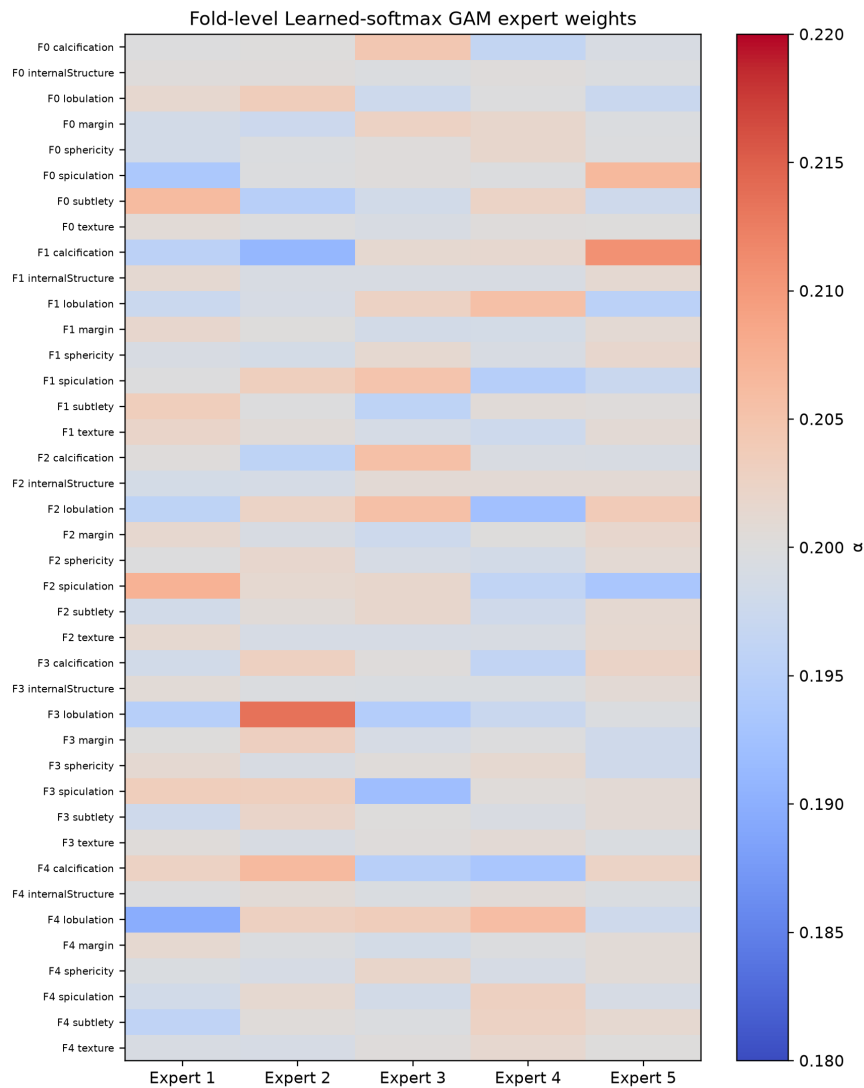
Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T16.csv.

Learned-softmax GAM adds a second WHY layer: five expert outputs per concept are mixed with nonnegative weights summing to one. Table RPT-T16 and Figure RPT-F11 show that the weights moved away from the uniform 0.2 initialization, although many movements are modest and fold-dependent. Learned mixtures therefore constitute evidence of optimisation, not proof that each expert represents a distinct clinical mechanism.



RPT-F10. Empirical OOF contribution profiles; descriptive presentation only, not causal shape functions.

What does this mean? Contribution decompositions make score construction auditable and permit case-level signed bars, but magnitude and sign remain properties of the trained decision function. They do not validate the underlying concepts or establish clinical causation. The private qualitative appendix pairs contribution bars with CT context and concept prediction/target evidence so WHY is not detached from WHAT and Prediction.



RPT-F11. Fold-level Learned-softmax GAM expert mixture weights.

6.5 Results — HOW

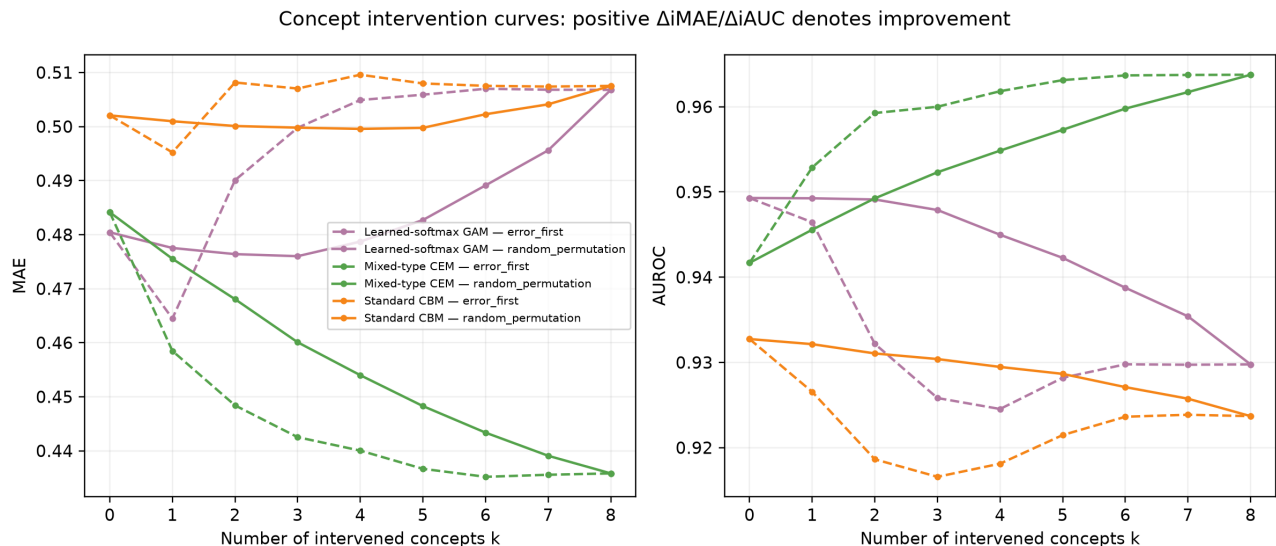
What was measured? Concept interventions replace 0...8 groups using the registered model-specific semantics. At every k, five-fold OOF predictions are pooled before calculating primary MAE and secondary AUROC. Random-permutation curves average 100 deterministic permutations per fold; error-first ordering ranks continuous absolute error or categorical total-variation distance without using the malignancy target.

RPT-T17. Concept-intervention summary

Model	Ordering	Baseline MAE	k=4 MAE	k=8 MAE	iMAE	Delta_iMAE	Baseline AUROC
Learned-softmax GAM	error_first	0.480	0.505	0.507	0.497	-0.016	0.949
Learned-softmax GAM	random_permutati on	0.480	0.479	0.507	0.484	-0.003	0.949
Mixed-type CEM	error_first	0.484	0.440	0.436	0.445	0.040	0.942
Mixed-type CEM	random_permutati on	0.484	0.454	0.436	0.456	0.028	0.942
Standard CBM	error_first	0.502	0.510	0.508	0.506	-0.004	0.933
Standard CBM	random_permutati on	0.502	0.500	0.508	0.501	0.001	0.933

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T17.csv.

What did we observe? Standard CBM showed little average MAE improvement under random ordering and deterioration under error-first integration. Mixed-type CEM and Learned-softmax GAM showed larger positive Delta_iMAE under random permutations, with secondary Delta_iAUC patterns that were also model-dependent. Table RPT-T17 retains baseline, intermediate, $k=8$, iMAE, Delta_iMAE, iAUC, and Delta_iAUC; Figure RPT-F12 displays all $k=0\dots 8$ curves.



RPT-F12. $k=0\dots 8$ concept-intervention curves under random and error-first orderings.

Error-first results were not uniformly better than random ordering. Correcting the currently worst-predicted concept first can expose compensating errors elsewhere in the model, and later interventions may reverse an early benefit. This is an important negative result because it shows that concept correction is not a monotonic repair operation.

What does this mean? HOW evidence tests the model's dependence on concept representations, not the causal effect of changing a patient's radiology. Positive Delta_iMAE and Delta_iAUC consistently denote improvement, but the magnitude depends on architecture, ordering, and metric. Case-level before/after intervention values were not persisted for RPT-FA06, so the private figure marks HOW as DATA_NOT_PERSISTED rather than fabricating a trajectory.

Scientific conclusion codes: INTERVENTION_BENEFIT_MODEL_DEPENDENT

6.6 Integrated Interpretation

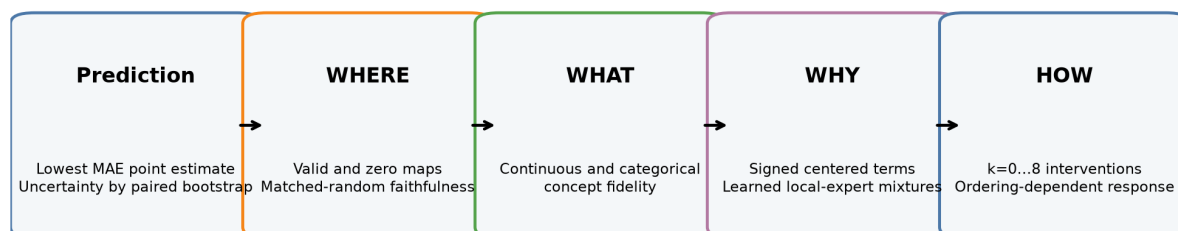
Prediction, WHERE, WHAT, WHY, and HOW answer different questions and should not be collapsed into a single explainability score. Prediction establishes task performance. WHERE tests spatial sensitivity but is limited by zero maps and weak matched-random advantages. WHAT measures whether named concepts match reader evidence. WHY exposes score composition. HOW probes whether correcting representations changes the output.

RPT-T18. WHERE-WHAT-WHY-HOW synthesis

Layer	Question	Main evidence	Boundary
Prediction	How accurately is malignancy scored?	Learned-softmax GAM has the lowest point-estimate MAE; paired support is model-dependent.	Radiologist assessment, not pathology.
WHERE	Where is the output spatially sensitive?	66,769 valid maps; saliency often did not exceed matched random masks.	6,955 post-ReLU zero maps; exact mechanism unavailable.
WHAT	Which concepts were predicted?	Continuous fidelity varied by concept; categorical hard-F1 was limited.	Categorical targets are reader-vote distributions.
WHY	How do concepts enter the score?	Signed centered terms and learned GAM mixtures reconstruct the score.	Centering constants are not importance; mean absolute aggregate was not persisted.
HOW	How does correcting concepts alter prediction?	Intervention benefit was strongest for CEM/GAM and ordering-dependent.	Interventions test model dependence, not causal clinical effects.

Full machine-readable rows and field-level Catalogue provenance are retained in RPT-T18.csv.

The strongest integrated interpretation belongs to a model only when these layers are read together. Learned-softmax GAM has the best primary point estimate and an auditable additive score, but its spatial and intervention evidence still carries limitations. Mixed-type CEM benefits substantially from intervention despite imperfect categorical fidelity. Standard CBM remains simple and traceable but does not gain a prediction advantage over Black-box.



Integrated conclusion: interpretability evidence is complementary, model-dependent, and bounded

RPT-F13. Integrated Prediction-WHERE-WHAT-WHY-HOW interpretation and its boundaries.

Table RPT-T18 and Figure RPT-F13 therefore present a chain of supported claims and boundaries rather than a winner-takes-all dashboard. The conclusion is that interpretability is multidimensional and model-dependent: a useful explanation must specify which layer it supports, what frozen evidence underlies it, and what it cannot establish.

7. Discussion

The primary result favours Learned-softmax GAM at the point-estimate level, and paired MAE comparisons support meaningful improvement over Black-box and Standard CBM. This suggests that explicit concept-local nonlinearities can improve continuous scoring while preserving additive decomposition. However, confidence intervals and secondary discrimination prevent an overly simple ranking: the best regression point estimate is not synonymous with statistically superior AUROC across every pair.

The explanation layers reveal trade-offs that prediction metrics alone cannot show. Concept models provide WHAT and WHY evidence unavailable to Black-box, yet concept accuracy varies and categorical hard-F1 remains limited. Intervention improvements in CEM and GAM indicate that imperfect predicted concepts constrain the task path, but deterioration in other settings shows that the model may exploit correlated or compensating concept errors.

Spatial evidence is the clearest caution. Thousands of legitimate post-ReLU zero maps and predominantly weak saliency-versus-random differences mean that visually appealing overlays should not dominate the scientific story. A Grad-CAM overlay is best treated as a local sensitivity view whose credibility is strengthened only when quantitative faithfulness and map-validity accounting agree.

Clinically, the framework offers a disciplined way to communicate model behaviour, not a diagnosis. Reader ratings encode radiological assessment and disagreement; they do not establish histopathological truth. External validation, calibration for deployment, prospective workflow testing, and pathology-linked outcomes would be required before any clinical claim.

8. Limitations

First, the target is a radiologist-assessed malignancy score rather than pathology-confirmed disease. The patient-grouped internal cross-validation design controls leakage but cannot establish transportability to another institution, scanner distribution, or clinical workflow. Only one preregistered seed per fold was used, so the reported bootstrap intervals describe patient-sampling uncertainty rather than training-seed variability.

Second, concept ground truth inherits reader variability. Continuous means compress disagreement, and categorical vote distributions can be sparse. Hard modal macro-F1 excludes true ties and is secondary to the full distributional metrics. Concept interventions replace internal representations with reader-derived targets; they should not be interpreted as feasible clinical manipulations or causal effects.

Third, Grad-CAM is a nodule-level spatial proxy. The 6,955 undefined maps are confirmed finite post-ReLU all-zero maps, but the exact pre-ReLU/gradient mechanism was not persisted. The observed concentration is therefore reported as `SYSTEMATIC_MODEL/TARGET_ISSUE`, not resolved into zero gradients, zero channel weights, or negative weighted sums without prohibited new forward passes.

Finally, some presentation goals are constrained by what was frozen. The model-by-concept mean absolute centered contribution aggregate and case-level intervention before/after trajectory were not persisted. The report explicitly marks these items `DATA_NOT_PERSISTED` and does not convert descriptive plotting or narrative memory into a new authoritative scientific result.

9. Conclusion

This unified comparison shows that prediction and explanation should be evaluated as a chain of distinct questions. Learned-softmax GAM achieved the lowest primary MAE point estimate, while paired uncertainty showed where this advantage was and was not decisive. Concept models added inspectable representations, signed score decompositions, and intervention experiments that a Black-box predictor cannot provide.

The explanation evidence also imposed meaningful limits. Concept fidelity varied, intervention benefit was not monotonic or universal, Grad-CAM maps could be undefined, and saliency masks often failed to outperform matched random masks. These are not peripheral audit details; they determine how strongly each explanation can be interpreted.

The resulting framework supports transparent research reporting of radiologist-assessed pulmonary-nodule malignancy. It does not establish pathology-level diagnosis, causal concepts, ground-truth localisation, or clinical readiness. Its central contribution is a reproducible evidence structure linking Prediction, WHERE, WHAT, WHY, and HOW while keeping each claim bound to its frozen source and interpretation boundary.

Public Reproducibility Appendix

All scientific values are read from the user-approved 2,395-item Results Catalogue and its registered frozen sources. The report-revision supplement binds the Catalogue registry SHA, Catalogue manifest SHA, and both approved planning-document SHAs. Section manifests and reverse-traceability rows connect every rendered table, figure, caption, and conclusion code to Catalogue item IDs and source hashes.

P5–P9 checkpoints, histories, predictions, metrics, evaluations, OOF rows, interventions, Grad-CAM maps, occlusion rows, and faithfulness payloads remain read-only. Report generation performs no training, model forward pass, test inference, bootstrap recomputation, or new scientific job. The private archive remains outside Git and stores full-resolution case assets under opaque CASE labels.

The archive contains 1,698 files and 14,386,651,621 bytes under a completed SHA-verified manifest. The six mandatory PDFs are rendered page by page with Poppler at 150 DPI, inspected through contact sheets and original-resolution pages, and checked with pypdf/pdfplumber for metadata, text, numbering, fonts, and page

integrity before P10 can enter Awaiting_User_Approval.

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